Numerical Evaluation of the Canonical System up to Weight 6

Regularization at $x \to 0$ and Adaptation of Czakon–Tancredi (arXiv:2606.30354)

1 Strategy of Czakon–Tancredi, and what simplifies here

Ref. [1] solves $d\vec{J} = \varepsilon\, A(\vec{x})\, \vec{J}$ numerically by integrating along a path, adjoining to the canonical system a set of auxiliary *rational* differential equations for every non-rational building block (square roots, elliptic periods, ...) that appears in A. This avoids ever having to implement the transcendental functions themselves: the enlarged system is purely rational and can be integrated with a standard adaptive solver.

In the present 371×371 system, the connection one-form has, along the line $y \equiv y_0 = 3/15$, only *one* algebraic building block, $r(x, y_0) = \sqrt{x\, y_0\, (1 - x - y_0)}$. Unlike an elliptic period, r is an ordinary algebraic function with a closed form on the real line for $0 < x < 1 - y_0$ (no differential equation needs to be adjoined for it: it is simply evaluated). The only genuine complication is that $x = 0$ is a regular singular (Fuchsian) point of the system, so the paper's "integrate directly" strategy must be preceded by the regularization worked out in Sections 2–4 below.

2 Setup

The canonical system is $D\vec{J} = \varepsilon\, \tilde{A}\, \vec{J}$, where $\tilde{A}(x, y)$ (`Atilde.m`) is a matrix of logarithms of alphabet letters, so that the genuine connection one-form is obtained by differentiating entrywise,

$$A_x^{\text{full}}(x, y) \equiv \frac{\partial \tilde{A}}{\partial x} = \texttt{dx.m}\,, \qquad A_y^{\text{full}}(x, y) \equiv \frac{\partial \tilde{A}}{\partial y} = \texttt{dy.m}\,. \tag{1}$$

Restricting to the line $y \equiv y_0 = 3/15$ and parametrizing $x = x(t)$, the system collapses to a genuine ODE in t,

$$\frac{d\vec{J}}{dt} = \varepsilon\, \dot{x}(t)\, A_x^{\text{full}}\big(x(t), y_0\big)\, \vec{J}(t)\,, \tag{2}$$

since $\dot{y} = 0$ along this path.

The point $x = 0$ is a Fuchsian singularity of A_x^{full}. Splitting off the pole exactly,

$$A_x^{\text{full}}(x, y_0) = \frac{A_x}{x} + B(x, y_0)\,, \qquad A_x \equiv \operatorname*{Res}_{x=0} A_x^{\text{full}}(x, y_0) = \texttt{toexp1.m}\,, \tag{3}$$

where $B(x, y_0) \equiv A_x^{\text{full}}(x, y_0) - A_x/x$ is regular (finite) as $x \to 0$. Here A_x is a constant 371×371 matrix, verified to coincide exactly with `toexp1.m` and with the residue of `dx.m` at $x = 0$.

3 Regularization at $x \to 0$

Define

$$M(x, \varepsilon) \equiv \exp\big(\varepsilon A_x \log x\big) = x^{\varepsilon A_x}\,, \tag{4}$$

which is exactly `exponentiated1.m` with $\delta \to x$, and the *regularized vector*

$$\vec{J}'(x) \equiv M(x,\varepsilon)^{-1}\vec{J}(x)\,, \qquad \text{equivalently} \qquad \vec{J}(x) = M(x,\varepsilon)\,\vec{J}'(x)\,. \tag{5}$$

Since $M(x,\varepsilon)$ is built purely from A_x, it commutes with A_x and

$$\frac{dM}{dx} = \frac{\varepsilon}{x}\,A_x\,M(x,\varepsilon) = \frac{\varepsilon}{x}\,M(x,\varepsilon)\,A_x\,. \tag{6}$$

Differentiating $\vec{J} = M\vec{J}'$ and imposing the original equation,

$$\frac{\varepsilon}{x}A_x M\vec{J}' + M\frac{d\vec{J}'}{dx} = \varepsilon\left(\frac{A_x}{x} + B(x,y_0)\right)M\vec{J}'\,. \tag{7}$$

The A_x/x terms cancel *identically* on both sides — this cancellation is exactly what fixes the sign of the exponent in $M(x,\varepsilon)$ to be $+\varepsilon A_x$, matching the convention already used to build `exponentiated1.m`; no sign flip is required. What remains is

$$M(x,\varepsilon)\,\frac{d\vec{J}'}{dx} = \varepsilon\,B(x,y_0)\,M(x,\varepsilon)\,\vec{J}'(x)\,, \tag{8}$$

i.e. the **exact, pole-free equation**

$$\boxed{\frac{d\vec{J}'}{dx} = \varepsilon\,M(x,\varepsilon)^{-1}B(x,y_0)\,M(x,\varepsilon)\,\vec{J}'(x)}\,, \qquad \frac{d\vec{J}'}{dt} = \dot{x}(t)\times(\text{same, at } x = x(t))\,, \tag{9}$$

with

$$B(x,y_0) = \texttt{dx.m}(x,y_0) - \frac{\texttt{toexp1.m}}{x} \tag{10}$$

manifestly finite (no pole) at $x = 0$. The conjugated coefficient $M(x,\varepsilon)^{-1}B(x,y_0)M(x,\varepsilon)$ remains bounded as $x \to 0$, order by order in ε: any $\log x$ dependence it carries integrates to terms like $x\log x \to 0$, never to a divergence. Hence this equation may be integrated numerically starting exactly at $x = 0$, using the known boundary value $\vec{J}'(0,y_0)$.

4 Recovering the physical vector

At the target point x_1,

$$\vec{J}(x_1,y_0) = M(x_1,\varepsilon)\,\vec{J}'(x_1,y_0) = \texttt{exponentiated1.m}\big|_{\delta=x_1}\cdot\vec{J}'(x_1,y_0)\,. \tag{11}$$

Practical note on $M(x,\varepsilon)^{-1}$. No new computation is required: since $M(x,\varepsilon)^{-1} = \exp(-\varepsilon A_x\log x)$ is built from the *same* spectral projectors and nilpotents already saved in `toJD/Npowers_⟨λ⟩_toexp1.m`, it is obtained by flipping the sign of ε (equivalently $\lambda \to -\lambda$ in $\delta^{\varepsilon\lambda}$ and $\varepsilon\log\delta \to -\varepsilon\log\delta$ in the nilpotent series).

5 Weight-by-weight (order-by-order in ε) reduction

Rather than solving (9) at a fixed numerical ε, we extract the Taylor coefficients in ε directly – this is both what is physically wanted (the finite remainder is built order by order in ε) and what makes the numerics cheap, since $M(x,\varepsilon)$ has an *exactly known*, finite eps-expansion built from already-computed exact matrices.

5.1 Expansion of $M(x,\varepsilon)$

Since $A_x =$ `toexp1.m` has nilpotency order ≤ 2 (only at $\lambda=-3$; order 1 at $\lambda=1$; semisimple elsewhere), for each eigenvalue λ

$$x^{\varepsilon\lambda}\Big(P_\lambda + \varepsilon\log x\, N_\lambda + \tfrac{1}{2}(\varepsilon\log x)^2 N_\lambda^2\Big) \tag{12}$$

is a product of the ordinary scalar exponential series $x^{\varepsilon\lambda}=\sum_n \frac{(\lambda\log x)^n}{n!}\varepsilon^n$ with a *polynomial* of degree ≤ 2 in ε. Convolving the two series and summing over λ gives, for any truncation order W,

$$M(x,\varepsilon)=\sum_{j=0}^{W-1}\varepsilon^j M_j(x)+O(\varepsilon^W)\,,\qquad M_j(x)=\sum_\lambda\sum_{k=0}^{\min(2,j)}\frac{(\lambda\log x)^{j-k}}{(j-k)!}\cdot\frac{(\log x)^k}{k!}\,N_\lambda^k\,. \tag{13}$$

Every $M_j(x)$ is an explicit *linear combination of the fixed, exactly-known matrices* $\{P_\lambda, N_\lambda, N_\lambda^2\}$ (already saved in `toJD/Npowers_`$\langle\lambda\rangle$`_toexp1.m`) with scalar coefficients that are simple powers of $\log x$. No matrix multiplication is needed to build $M_j(x)$ – only scaled matrix additions. In fact, since $(\log x)^{j-k}(\log x)^k=(\log x)^j$ regardless of k, this simplifies further to $M_j(x)=(\log x)^j\,Q_j$ with

$$Q_j\equiv\sum_\lambda\sum_{k=0}^{\min(2,j)}\frac{\lambda^{j-k}}{(j-k)!\,k!}\,N_\lambda^k \tag{14}$$

a fixed, *x-independent* exact rational matrix, computable once ($Q_0=\sum_\lambda P_\lambda = \mathbb{1}$, $Q_1=\sum_\lambda(\lambda P_\lambda+N_\lambda)=A_x$, verified explicitly).

5.2 Expansion of $M(x,\varepsilon)^{-1}$

Since $M(x,\varepsilon)^{-1}=\exp(-\varepsilon A_x\log x)=M(x,-\varepsilon)$,

$$M(x,\varepsilon)^{-1}=\sum_{j=0}^{W-1}\varepsilon^j\,(-1)^j M_j(x)+O(\varepsilon^W)\,, \tag{15}$$

i.e. the inverse's coefficients are obtained from (13) by a sign flip on odd j – no independent computation is required.

5.3 Expansion of the conjugated matrix and the hierarchy for $\vec{J}\,'$

Write $C(x,\varepsilon)=M(x,\varepsilon)^{-1}B(x,y_0)M(x,\varepsilon)=\sum_{j\ge0}\varepsilon^j C_j(x)$, with

$$C_j(x)=\sum_{p+q=j}(-1)^p M_p(x)\,B(x,y_0)\,M_q(x)=(\log x)^j\sum_{p=0}^{j}(-1)^p Q_p\,B(x,y_0)\,Q_{j-p}\,, \tag{16}$$

using $M_p(x)M_q(x)\propto(\log x)^{p+q}=(\log x)^j$ from the previous subsection. Expanding $\vec{J}\,'(x,\varepsilon)=\sum_{k\ge0}\varepsilon^k\vec{J}\,'^{(k)}(x)$ and matching powers of ε in (9) gives the **exact** recursive hierarchy

$$\boxed{\frac{d\vec{J}\,'^{(0)}}{dx}=0\,,\qquad \frac{d\vec{J}\,'^{(m)}}{dx}=\sum_{k=0}^{m-1}C_{m-1-k}(x)\,\vec{J}\,'^{(k)}(x)\,,\quad m=1,\dots,W-1} \tag{17}$$

Crucially, the right-hand side at order m involves *only* $\vec{J}'^{(0)}, \ldots, \vec{J}'^{(m-1)}$, never $\vec{J}'^{(m)}$ itself. The system is therefore *block lower-triangular*: $\vec{J}'^{(0)}$ is the constant weight-0 boundary vector (rational, valid everywhere – consistent with `sol0.m`), and each subsequent order is obtained by integrating a source built entirely from lower orders. This is precisely the iterated-integral structure of the canonical system, made explicit rather than left implicit in a black-box ε-integration.

5.4 Practical evaluation of $B(x, y_0)$

$B(x, y_0) = \texttt{dx.m}(x, y_0) - A_x/x$ is ε-independent. Along the fixed line $y = y_0$, its only non-rational content is the algebraic letter $r(x, y_0) = \sqrt{x\, y_0(1 - x - y_0)}$ and the derived combinations $r_1 = r(x-1)/y_0$, $r_2 = r(x+y_0)/y_0$ appearing in `dx.m`; substituting these closed forms turns $B(x, y_0)$ into an explicit (compilable) function of x alone. No auxiliary ODE is needed for r on this path, since it is not a period of a higher-genus curve but an elementary algebraic function.

6 Numerical integration

The state vector is the stack $\left(\vec{J}'^{(1)}, \ldots, \vec{J}'^{(W-1)}\right) \in \mathbb{R}^{371(W-1)}$ ($\vec{J}'^{(0)}$ is constant and known, so it is not part of the dynamical state). Each block evolves via (17), with the $C_j(x)$ evaluated on the fly at each requested x (via Q_j, cheap scaled sums, plus the compiled $B(x, y_0)$ and 371×371 matrix products per C_j). This is integrated with `NDSolve` (adaptive Runge–Kutta by default), starting from a small $x = x_{\text{start}} \approx 0$ (a numerical regulator for $\log x$; $\vec{J}'$ is analytic at $x = 0$ so $\vec{J}'(x_{\text{start}}) \simeq \vec{J}'(0)$ to excellent approximation) up to the target point x_1. Finally, the physical vector at x_1 is recovered by re-multiplying and re-expanding in ε up to $O(\varepsilon^{W-1})$:

$$\vec{J}(x_1, y_0) = M(x_1, \varepsilon)\, \vec{J}'(x_1, \varepsilon)\Big|_{\varepsilon\text{-expanded to order } W-1} = \sum_{m=0}^{W-1} \varepsilon^m \sum_{k=0}^{m} M_{m-k}(x_1)\, \vec{J}'^{(k)}(x_1)\,. \tag{18}$$

This is implemented in `integrate_canonical.wl`, parameterized by the target point x_1 and the maximal weight $W - 1 \leq 6$.

References

[1] M. Czakon and L. Tancredi, *Solution of Canonical Differential Equations for Integrals on Arbitrary Geometries*, arXiv:2606.30354 [hep-ph].